

Real-Time Spatial Digital Twin of a Beef Cattle Production System

A real-time 3D twin of a beef cattle facility in Unreal Engine 5, per-agent health and behavior states, temporal replay, and a swap-ready pipeline for real sensor integration

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01 The Problem

Beef cattle production couples biological and engineered subsystems, yet workflows still run on **disconnected spreadsheets, passive sensor logs, and informal spatial records**. No unified platform lets managers test interventions **spatially, before real-world rollout**. Existing precision-livestock tools solve narrow subproblems; none unifies physiological state, spatial behavior, environmental variability, and decision support in one real-time interactive environment.

02 Five-Layer Architecture

I5	Immersive Visualization UE5 real-time rendering · KPI dashboard · heatmap · minimap · multi-camera views
I4	Simulation & Decision Support Behavior Trees · NavMesh pathfinding
I3	State Management MQTT broker + UE5 Blueprint Event Dispatchers
I2	Data Acquisition Python ETL · timestamp-aligned JSON · TAMU/feedlot/penID/animalID/datatype
I1	Physical Representation Georeferenced 3D facility scene · real-time LOD · FBX asset pipeline

03 Prior Work Gaps

No published livestock digital twin unified per-animal health state, real-time messaging, and immersive visualization in one system. Jeong et al. added environmental sensing but no animal-level representation. Neethirajan & Kemp and Mallinger et al. made the conceptual case without a working implementation. This system fills that gap.

04 The Live Digital Twin



FIG 1 Live Unreal Engine 5 operator interface. Pen B active, agent TAG_1012 INFECTED at HLI 92.7. Health/behavior state donuts, per-agent telemetry, movement heatmap, and timeline scrubber.

05 Key Results

- ✓ **Five layers run end to end**, from a georeferenced 3D facility to a live operator dashboard.
- ✓ **Any moment is replayable**: rewind an episode to its first case and watch it unfold.
- ✓ **Stable through dropouts** via last-value carry-forward, with no crashes or logic errors.
- ✓ **At-risk agents surface early**, flagged from rising exposure before any clinical onset.
- ✓ **The data source swaps out** with zero engine changes, confirmed with a mock live stream.
- ✓ **Runs on one workstation laptop**, no cloud or render cluster needed.

06 Why It Matters

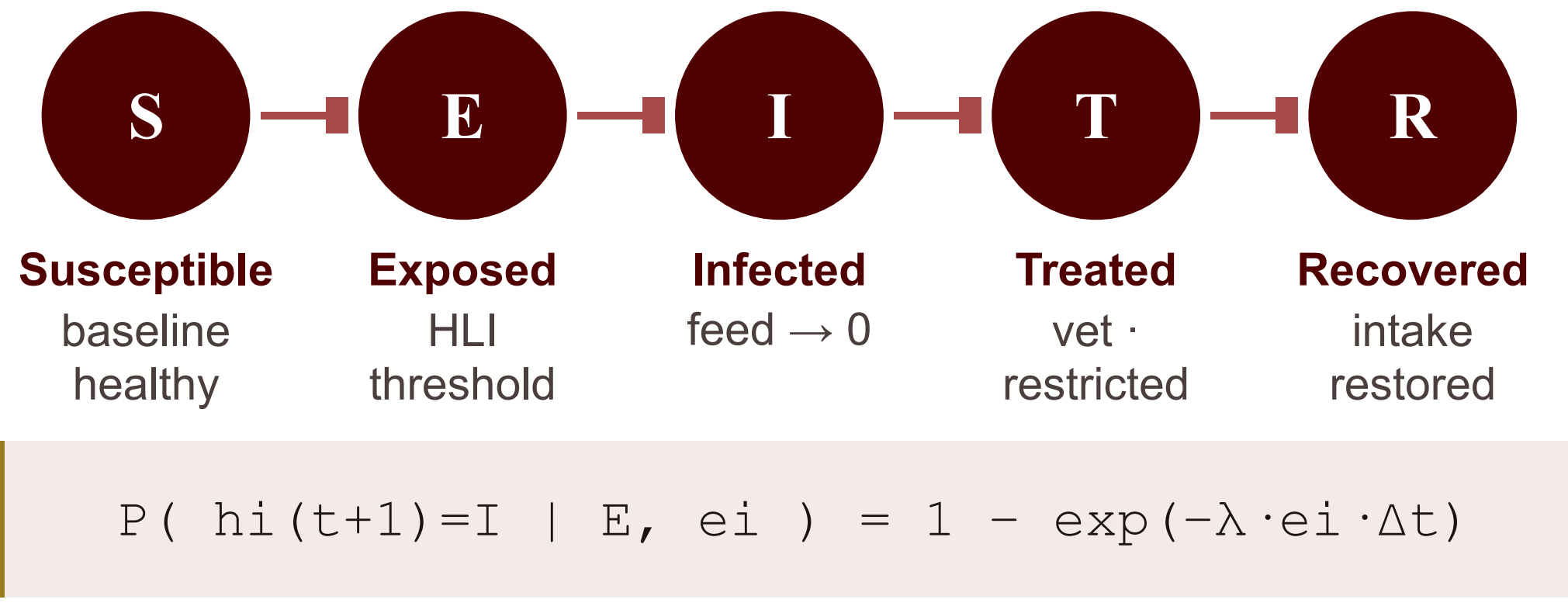
Per-agent state State vectors formalize a modeling standard most deployed livestock platforms lack; they operate only at herd-level aggregates.	Treatment as state SEIRT elevates Treatment to an explicit discrete state, whose timing and efficacy drive welfare and economic outcome.	Formal layer boundary The five layers separate what the system is, reports, knows, infers, and displays, enabling layer-by-layer validation.
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07 Per-Agent State Model

$$\mathbf{si}(t) = [\mathbf{hi}, \mathbf{pi}, \mathbf{bi}, \mathbf{ei}]$$

- $\mathbf{hi}(t)$ Health: discrete SEIRT state
- $\mathbf{pi}(t)$ Position: 3D spatial coordinate
- $\mathbf{bi}(t)$ Behavior: feeding · resting · walking · drinking · social clustering
- $\mathbf{ei}(t)$ Exposure: local environment index

08 SEIRT Health Pipeline



09 Reading the Interface

- HLI / THI** Live environmental stream values that trigger agent state changes when thresholds are crossed.
- Donuts** Health and behavior state distributions across the active pen, updated every tick.
- Agent panel** Full state vector for the selected animal: health, behavior, weight, feed and water intake, stress score.
- Scrubber** Replays any recorded moment; all views update in sync as you move it.
- Alerts** Triggered when an agent transitions to Infected state, notifying the operator for intervention.
- Pen Views** Switch between pens to see live agent positions and color-coded health states on heatmap
- Heatmap** Spatial overlay of movement density for selected pen zone with health distribution, updated live from positions
- Facility View** top-down orthographic of the full facility, all three pens visible simultaneously

10 Future Work

Live RFID / IoT streams	ML predictive health models
Commercial-scale benchmarking	Producer & vet usability studies
Economic performance modules	CV behavioral detection

Synthetic Telemetry Corpus	85k+ RECORDS	30 DAYS	15 min INTERVAL	2880 TIMESTAMPS	30 AGENTS	3 PENS
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